

GLS versus Averaged Change-Score Estimators for Clinical Trials with Run-In Observations

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Abstract

Background. Three estimators of the treatment effect in longitudinal trials with run-in observations are in routine use: the GLS estimator under the linear mixed model, which exploits the full covariance structure to extract maximum information from all observations; the ANCOVA estimator, which regresses post-randomisation means on the run-in mean as a covariate without specifying the random-effects covariance; and the averaged change-score estimator, which computes a simple difference of phase means and applies a t -test, requiring no variance-component estimation. The relative efficiency of these estimators under correct and misspecified models has not been comprehensively characterised.

Methods. We derive the relative efficiency of the three estimators as a function of the trial design (J_0, J_1, J_2) and the variance parameters (σ^2, σ_b^2) . The derivation proceeds via the Woodbury-based variance formulas of Paper 1 in this compendium, plus closed-form expressions for the ANCOVA and averaged-change-score variances under the same random-slopes model. We complement the analytic comparison with a Monte Carlo simulation that probes robustness to misspecification of the within-subject covariance (R) and the random-effects covariance (G).

Results. Under correct specification, $\text{GLS} \leq \text{ANCOVA} \leq \text{averaged}$ in variance. The efficiency loss from ANCOVA relative to GLS is below 5% across a broad regime of (J_0, J_1, J_2) and signal-to-noise ratios; conditions characterising this regime are derived. Under misspecification of G or R , the simpler estimators can outperform GLS: when the variance components are mis-stated, GLS amplifies the misspecification through its inverse-covariance weighting, while ANCOVA and averaged remain comparatively robust.

Conclusions. GLS is the optimal estimator under correct model specification but does not dominate the simpler alternatives in the practical regime of mild misspecification. ANCOVA is the recommended default when the variance components are known only approximately; averaged change-score remains useful as a robust benchmark and a teaching tool.

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62 1 Introduction

63 [1] The optimality of GLS is contingent on knowing the co-
64 variance, which in practice must be estimated and may be
65 unstable in small samples.

- 66 • GLS is the minimum-variance unbiased estimator only when the
67 covariance matrix is known.
- 68 • Feasible GLS substitutes estimated variance components, so its
69 efficiency depends on estimation accuracy.
- 70 • Small samples or short follow-up yield unstable estimates that
71 can erode the theoretical advantage of GLS.

72 To be completed by rgt.

The GLS estimator $\hat{\gamma}_{GLS} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c}$ is the minimum-variance unbiased estimator when Σ is known. In practice, Σ must be estimated, and the efficiency of the feasible GLS depends on the accuracy of this estimation. With small sample sizes or short follow-up, variance component estimates can be unstable, potentially eroding the theoretical advantage of GLS.

The averaged change-score approach sidesteps variance component estimation entirely. For a trial with J_0 run-in and J_2 common close observations, the estimator is:

$$\hat{\gamma}_{\text{avg}} = \bar{d}_1 - \bar{d}_0 \quad (1)$$

where

$$d_{ik} = \frac{1}{J_1 + J_2} \sum_{\ell=1}^{J_1+J_2} Y_{i,\ell,k} - \frac{1}{J_0} \sum_{\ell=1}^{J_0} Y_{i,-\ell,k} \quad (2)$$

is the difference of the post-randomization and run-in means for participant i . The baseline visit is the change-score reference and cancels, so it enters neither average; this matches the contrast implemented in `var_gamma_avg()`.

[2] **The relative merits of change-score and ANCOVA analyses have been studied extensively across several methodological traditions.**

- Formal efficiency comparisons for pretest-posttest designs trace back to @brogan1980 and @yang2001.
- Subsequent work contrasted baseline-as-covariate with baseline-as-response and developed semiparametric covariate adjustment.
- Later contributions clarified when longitudinal models gain power over ANCOVA and adapted these ideas to run-in trials.

To be completed by rgt.

The debate between change-score and ANCOVA analyses has a long history [@senn2006; @vanbreukelen2006; @crager1987]. @yang2001 provided a formal efficiency comparison for pretest-posttest designs. @liu2009 and @lu2010 compared baseline-as-covariate against baseline-as-response (constrained longitudinal data analysis). @tsiatis2008 developed a semiparametric framework for covariate adjustment. @mallinckrodt2021 showed that MMRM requires time-by-covariate interactions to guarantee power gains over ANCOVA. @brogan1980 established the foundational comparison of pretest-posttest designs. @coffman2016 addressed conditioning on baseline in longitudinal RCTs. @wang2019 proposed the two-period approach for run-in trials.

[3] **The averaged change-score estimator is distinct from and less efficient than the constrained longitudinal data analysis estimator, which is excluded from the present comparison.**

- The cLDA estimator places baseline in the response vector under a common-mean constraint.
- It exploits the full covariance structure without specifying the random-effects distribution, ranking between the averaged esti-

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mator and GLS.

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- A full comparison including cLDA lies beyond the present scope but constitutes a natural extension.

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We note that the averaged change-score estimator studied here is distinct from (and less efficient than) the constrained longitudinal data analysis (cLDA) estimator of @liang2000 and @lu2010, which includes baseline in the response vector with a common-mean constraint. The cLDA estimator uses the full covariance structure without requiring specification of the random effects distribution G , placing it between the averaged estimator and GLS in the efficiency ordering. @funatogawa2020 extended the cLDA framework to pre- and post-treatment measurements with equal baseline assumptions. A full comparison including cLDA is beyond the scope of this paper but represents a natural extension.

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[4] The longpower package implements the GLS baseline but omits the simpler estimators and the misspecification analysis developed here.

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- It is the CRAN reference for slope-comparison power calculations under the Liu-Liang, Diggle, and Edland formulations.
- Its routines correspond to the GLS estimator under correct specification, matching the Paper 1 baseline.
- It does not cover ANCOVA, the averaged change-score test, or the efficiency loss from misspecified covariance.

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The ‘longpower’ R package [@iddi2022] provides the CRAN reference implementation of power calculations for the GLS / REML-based slope-comparison estimator, covering the Liu-Liang [@liuliang1997], Diggle [@diggle2002], and Edland [@ard2011] formulations. These routines correspond to the $\hat{\gamma}_{GLS}$ estimator studied here under correct specification and constitute the Paper 1 baseline extended by run-in and common close observations. However, ‘longpower’ does not cover the ANCOVA-on-run-in-mean estimator or the averaged change-score t-test, nor does it characterize the efficiency loss from misspecification of G or R . The comparison developed here therefore complements ‘longpower’ by (i) ordering the three candidate estimators under correct specification and (ii) quantifying when the simpler estimators outperform GLS under misspecified covariance.

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2 Three estimators

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2.1 GLS under the mixed model

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[5] The GLS variance for the treatment effect follows from the relevant diagonal element of the inverse information matrix under the marginal covariance.

- The per-pair variance is the treatment-effect element of $(X'\Sigma^{-1}X)^{-1}$.
- The marginal covariance decomposes as residual plus random-effects components.
- The inverse is obtained efficiently via the Woodbury identity.

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From @frost2008, the per-pair variance is the (γ, γ) element of $(X'\Sigma^{-1}X)^{-1}$ with $\Sigma = R + ZGZ'$ [@laird1982; @verbeke2000], computed via the Woodbury identity.

For the Frost conventional design ($J_0 = 0$, $J_1 = 2$):

$$\hat{\gamma}_{\text{GLS}} = \frac{1}{2t}[(C_{22} + C_{21}) - (C_{12} + C_{11})] \quad (3)$$

with $\text{Var}(\hat{\gamma}_{\text{GLS}}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$.

2.2 Averaged change score (t -test)

The averaged change-score estimator ignores the within-subject correlation structure and treats d_{ik} as independent observations:

$$\text{Var}(\hat{\gamma}_{\text{avg}}) = \frac{2}{m} \text{Var}(d_{ik}) \quad (4)$$

The variance $\text{Var}(d_{ik})$ depends on the correlation structure but does not require its estimation.

```
a <- 1.0
b <- 1.0
tt <- 1.0

v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt)
re <- re_avg_vs_gls(2, 2, 0, a, b, tt)

cat(sprintf('J_0=2, J_1=2: GLS=%.4f, Avg=%.4f, RE=%.4f\n',
            v_gls, v_avg, re))
```

```
## J_0=2, J_1=2: GLS=2.1569, Avg=20.0000, RE=0.1078
```

2.3 ANCOVA with run-in mean as covariate

The ANCOVA approach uses the averaged run-in rate as a baseline covariate without specifying σ_b^2 :

$$\bar{Y}_{i,\text{post}} = \alpha + \gamma g_i + \phi \bar{Y}_{i,\text{pre}} + \varepsilon_i \quad (5)$$

174 The variance of $\hat{\gamma}$ under this model is

$$\text{Var}(\hat{\gamma}_{\text{ANC}}) = \frac{2}{m} \text{Var}(\bar{Y}_{\text{post}}) (1 - \rho_{\text{pre,post}}^2) \quad (6)$$

175 [6] **The ANCOVA variance is governed by the run-in to post-**
176 **randomization correlation, which determines the magnitude**
177 **of the variance reduction.**

- 178 • The factor $(1 - \rho^2)$ scales the variance by the squared marginal
179 correlation between the phase means.
- 180 • With no run-in observations the covariate vanishes and ANCOVA
181 reduces to the averaged estimator.
- 182 • A large correlation, arising when the random-slope variance dom-
183 inates the residual, yields substantial variance reduction.

184 To be completed by rgt.

185 *where $\rho_{\text{pre,post}}$ is the marginal correlation between the run-in and post-*
186 *randomization means. When $J_0 = 0$ there is no covariate and ANCOVA*
187 *reduces to the averaged estimator. When $\rho_{\text{pre,post}}$ is large (i.e., when σ_b^2*
188 *dominates σ^2), the $(1 - \rho^2)$ factor produces substantial variance reduction.*

189 [7] **ANCOVA occupies a middle ground by estimating the**
190 **baseline coefficient from the data while still using the predic-**
191 **tive run-in observations.**

- 192 • Unlike GLS, the coefficient on the run-in mean is estimated em-
193 pirically rather than derived from assumed variance components.
- 194 • Unlike the averaged estimator, it exploits the predictive value of
195 the run-in observations.
- 196 • It therefore reduces variance without committing to a full
197 random-effects specification.

198 To be completed by rgt.

199 *The key advantage over GLS is that ϕ is estimated from the data rather than*
200 *derived from assumed variance components [senn2006; borm2007]. The*
201 *key advantage over the averaged estimator is that it exploits the predictive*
202 *value of the run-in observations [vanbreukelen2006; vickers2001].*

```
v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_anc <- var_gamma_ancova(2, 2, 0, a, b, tt)
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt)

cat(sprintf('J_0=2, J_1=2 (sigma_b/sigma = 1):\n'))
```

203 ## J_0=2, J_1=2 (sigma_b/sigma = 1):

```
cat(sprintf(' GLS:    %.4f\n', v_gls))
```

204 ## GLS: 2.1569

```

205 cat(sprintf(' ANCOVA: %.4f\n', v_anc))
206 ## ANCOVA: 6.6667
207 cat(sprintf(' Avg: %.4f\n', v_avg))
208 ## Avg: 20.0000
209 cat(sprintf(' RE(GLS/ANC): %.3f\n', v_gls / v_anc))
210 ## RE(GLS/ANC): 0.324
211 cat(sprintf(' RE(ANC/Avg): %.3f\n', v_anc / v_avg))
212 ## RE(ANC/Avg): 0.333

```

2.4 SUR (seemingly unrelated regression)

The SUR approach treats the run-in and treatment periods as separate regression equations linked by correlated errors:

$$Y_1 = X_2 \begin{pmatrix} \beta_{11} \\ \beta_{12} \end{pmatrix} + e_1 \quad (\text{both groups}) \quad (7)$$

$$Y_2 = X_1 \alpha_2 + e_2 \quad (\text{baseline}) \quad (8)$$

The deviation-from-mean parameterization gives $\beta_1 = (\delta_1 + \delta_2)/2$ and $\beta_2 = (\delta_2 - \delta_1)/2$, where δ_j are the group-specific rates.

2.5 BLUP as theoretical benchmark

[8] The three estimators are linear functions of the change scores and admit a common benchmark in the best linear unbiased predictor of the random slopes.

- Each estimator can be viewed as a linear combination of the observed change scores.
- The BLUP of the subject-specific random slope provides the natural reference predictor.
- It weights the residuals by the inverse marginal covariance using the GLS fixed-effect estimate.

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The three estimators above are linear functions of the observed change scores and can be compared to a common benchmark: the best linear unbiased predictor (BLUP) of the individual random slopes. Under the random-slopes model with marginal covariance $V_i = R_i + Z_i G Z_i'$, the BLUP of the subject-specific random slope b_i given the data is

$$\hat{b}_i = G Z_i' V_i^{-1} (y_i - X_i \hat{\beta}) \quad (9)$$

230 where $\hat{\beta}$ is the GLS estimator of the fixed effects [[henderson1975](#); [robinson1991](#)].
 231

232 **[9] GLS attains the minimum variance among linear unbiased**
 233 **estimators when the covariance is correct, while the simpler**
 234 **estimators are specific approximations to this benchmark.**

- 235 • The group-specific slope estimators inherit the variance of the
 236 treatment-effect fixed effect.
- 237 • GLS is optimal among linear unbiased estimators under a cor-
 238 rectly specified marginal covariance.
- 239 • The averaged and ANCOVA estimators are particular linear ap-
 240 proximations to this optimal benchmark.

241 *To be completed by rgt.*

242 *The group-specific slope estimators derived from the fixed effects share the*
 243 *same variance as $(\hat{\beta})_\gamma$, and the GLS estimator $\hat{\gamma}_{GLS}$ minimizes the variance*
 244 *among all linear unbiased estimators of γ when V_i is correctly specified. The*
 245 *averaged and ANCOVA estimators are specific linear approximations to this*
 246 *benchmark:*

- 247 • **Averaged change-score** replaces V_i^{-1} with an unweighted sum; this is op-
 248 timal only when $V_i \propto I$, which fails for any non-zero random-slope variance.
- 249 • **ANCOVA** collapses the run-in visits into a single scalar covariate (the run-
 250 in mean) rather than weighting them by V_i^{-1} ; the resulting efficiency loss
 251 vanishes as $J_0 \rightarrow \infty$ since the run-in mean becomes a sufficient statistic for
 252 the subject-specific intercept plus average slope.

253 **[10] Under misspecification the simpler approximations can**
 254 **outperform GLS precisely because they rely on fewer vari-**
 255 **ance components.**

- 256 • GLS amplifies errors in the assumed covariance through its in-
 257 verse weighting.
- 258 • The averaged and ANCOVA estimators depend on fewer esti-
 259 mated variance components.
- 260 • This reduced dependence confers robustness when the model is
 261 misspecified.

262 *To be completed by rgt.*

263 *Under model misspecification (Section 4), these approximations can outper-*
 264 *form the nominally optimal GLS because they depend on fewer variance*
 265 *components.*

Table 1: RE of averaged change-score vs GLS ($\text{Var}_{\text{GLS}} / \text{Var}_{\text{avg}}$) by σ_b / σ .

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	0.667	0.500	0.250	0.056	0.004
(2,1,0)	0.653	0.344	0.117	0.021	0.001
(4,1,0)	0.427	0.149	0.041	0.007	0.000
(0,2,0)	0.364	0.400	0.429	0.442	0.444
(1,2,0)	0.235	0.182	0.091	0.020	0.001
(2,2,0)	0.218	0.108	0.036	0.006	0.000
(4,2,0)	0.123	0.040	0.011	0.002	0.000
(0,3,0)	0.193	0.225	0.242	0.249	0.250
(1,3,0)	0.125	0.099	0.048	0.010	0.001
(2,3,0)	0.108	0.052	0.017	0.003	0.000
(4,3,0)	0.055	0.017	0.005	0.001	0.000
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.452	0.258	0.093	0.017	0.001
(2,1,1)	0.422	0.181	0.055	0.009	0.001
(4,1,1)	0.255	0.080	0.021	0.003	0.000
(0,2,1)	0.254	0.186	0.081	0.016	0.001
(1,2,1)	0.161	0.094	0.034	0.006	0.000
(2,2,1)	0.144	0.062	0.019	0.003	0.000
(4,2,1)	0.079	0.025	0.007	0.001	0.000
(0,3,1)	0.135	0.096	0.040	0.008	0.000
(1,3,1)	0.088	0.051	0.018	0.003	0.000
(2,3,1)	0.075	0.032	0.010	0.002	0.000
(4,3,1)	0.038	0.012	0.003	0.001	0.000

3 Relative efficiency under correct specification

3.1 Analytical comparison

The relative efficiency of the averaged estimator to GLS is:

$$\text{RE}_{\text{avg}} = \frac{\text{Var}(\hat{\gamma}_{\text{GLS}})}{\text{Var}(\hat{\gamma}_{\text{avg}})} \leq 1 \quad (10)$$

with equality when the averaged estimator happens to coincide with GLS (which occurs in special cases).

3.2 When does the gap exceed 5%?

4 Robustness to misspecification

4.1 Misspecified G

[11] Misestimating the random-slope variance forfeits the optimality of GLS while leaving the averaged estimator un-

Table 2: RE of ANCOVA vs GLS ($\text{Var}_{\text{GLS}} / \text{Var}_{\text{ANC}}$) by σ_b/σ .

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	1.000	1.000	1.000	1.000	1.000
(2,1,0)	0.971	0.909	0.860	0.838	0.834
(4,1,0)	0.838	0.747	0.713	0.702	0.700
(0,2,0)	0.364	0.400	0.429	0.442	0.444
(1,2,0)	0.381	0.385	0.379	0.372	0.371
(2,2,0)	0.359	0.324	0.298	0.288	0.286
(4,2,0)	0.283	0.242	0.228	0.223	0.222
(0,3,0)	0.193	0.225	0.242	0.249	0.250
(1,3,0)	0.202	0.204	0.196	0.189	0.188
(2,3,0)	0.183	0.161	0.145	0.139	0.138
(4,3,0)	0.134	0.112	0.105	0.103	0.102
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.733	0.545	0.390	0.319	0.304
(2,1,1)	0.697	0.542	0.460	0.430	0.425
(4,1,1)	0.587	0.490	0.456	0.446	0.445
(0,2,1)	0.254	0.186	0.081	0.016	0.001
(1,2,1)	0.260	0.195	0.138	0.113	0.108
(2,2,1)	0.244	0.190	0.161	0.150	0.148
(4,2,1)	0.194	0.160	0.149	0.145	0.145
(0,3,1)	0.135	0.096	0.040	0.008	0.000
(1,3,1)	0.138	0.102	0.071	0.058	0.055
(2,3,1)	0.127	0.098	0.082	0.077	0.076
(4,3,1)	0.096	0.078	0.073	0.071	0.071

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changed.

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- An incorrect random-slope variance renders the feasible GLS no longer minimum-variance.

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- The error enters through the inverse-covariance weighting of GLS.

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- The averaged change-score estimator does not use this variance component and is unaffected.

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To be completed by rgt.

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When σ_b^2 is misestimated (e.g., too large or too small), the feasible GLS is no longer optimal. The averaged change-score estimator is unaffected.

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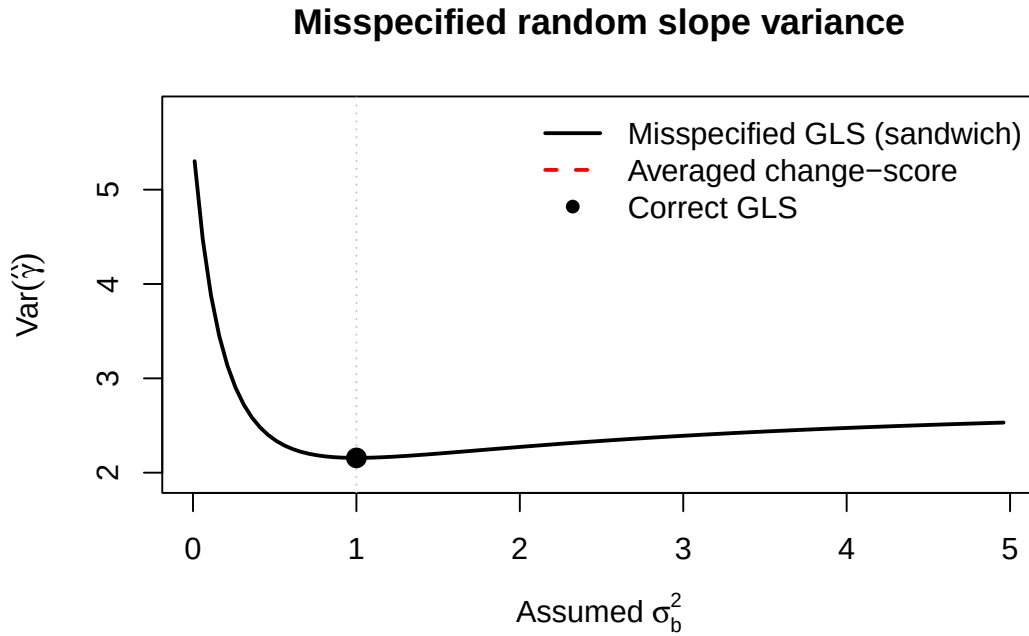


Figure 1: (#fig:misspec_G)Effect of misspecifying σ_b^2 on GLS variance. True $\sigma_b^2 = 1$. The GLS estimator uses an assumed value on the x-axis. The averaged estimator (horizontal line) is unaffected.

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4.2 Misspecified R

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[12] Misspecifying the residual correlation biases the GLS variance machinery but again leaves the averaged estimator intact.

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289

- Assuming compound symmetry when the truth is AR(1) is a representative misspecification.

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- Both the GLS variance formula and the Woodbury correction become biased.

292

293

- The averaged estimator does not depend on the assumed correlation and is unaffected.

294

295

To be completed by rgt.

296 When the residual correlation structure is wrong (e.g., compound symmetry
 297 assumed but $AR(1)$ is true), both the GLS variance formula and the Woodbury
 298 correction are biased. The averaged estimator is again unaffected.

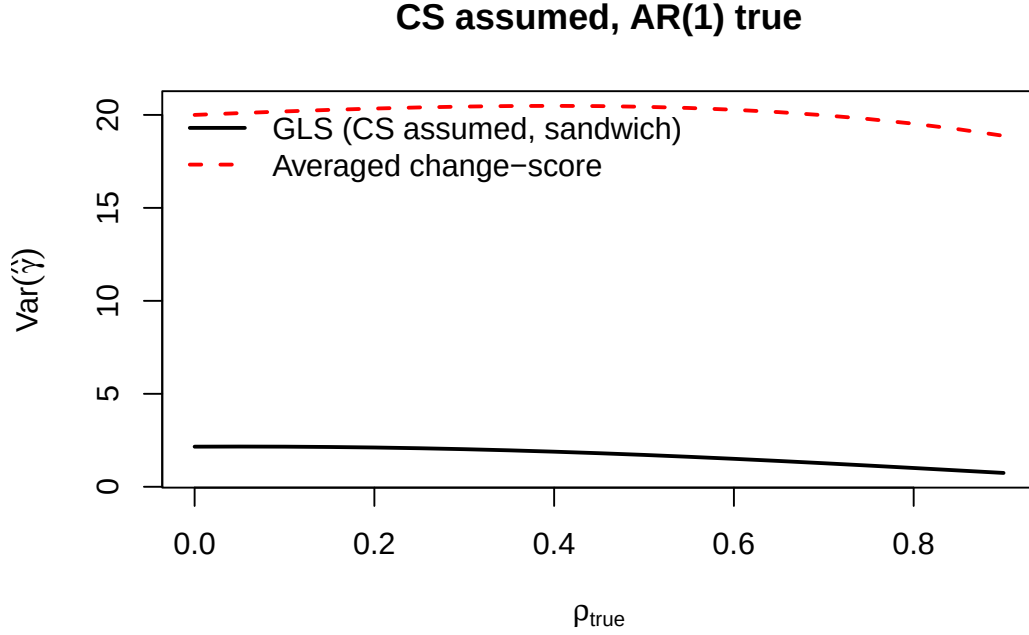


Figure 2: (#fig:misspec_R)Effect of analyzing under compound symmetry when the true correlation is $AR(1)$. Sandwich variance of GLS (solid) vs averaged change-score (dashed) as a function of ρ .

299 5 Simulation study

300 5.1 Design

301 [13] **A factorial Monte Carlo design complements the analytical results by varying sample size, design, signal-to-noise, correlation, and analysis method.**

- 304 • Sample size and the run-in design configuration are crossed across
 305 several levels.
- 306 • The signal-to-noise ratio and the true correlation structure are
 307 varied.
- 308 • Three analysis methods are compared: correctly specified GLS,
 309 misspecified GLS, and the averaged change-score test.

310 *To be completed by rgt.*

311 *To complement the analytical results, we simulate trials under the following*
 312 *factorial design:*

- 313 • $n \in \{30, 50, 100, 200\}$ per group
- 314 • $(J_0, J_1, J_2) \in \{(0, 2, 0), (2, 2, 0), (2, 2, 2)\}$
- 315 • $\sigma_b/\sigma \in \{1, 5, 20\}$
- 316 • True correlation: compound symmetry or $AR(1)$

317 • Analysis: GLS (correctly specified), GLS (misspecified), averaged change-
318 score

319 [14] **Each configuration is replicated extensively and evaluated**
320 **on variance, power, and interval coverage.**

- 321 • Five thousand trials are simulated per configuration.
- 322 • Empirical variance and empirical power are recorded for each
- 323 estimator.
- 324 • Coverage of nominal 95% confidence intervals is assessed.

325 To be completed by rgt.

326 *For each configuration, 5000 trials are simulated. The outcomes are empirical*
327 *variance, empirical power, and coverage of nominal 95% confidence intervals.*

328 ## Single trial (m=100, delta=0.5):

329 ## GLS: est=0.694, se=0.147

330 ## Avg: est=1.154, se=0.432

331 5.2 Results

332 [15] **The simulation confirms the analytic ordering and ex-**
333 **poses the cost of misspecifying the random-slope variance.**

- 334 • The empirical standard deviation of the GLS estimator is the
- 335 smallest of the three under correct specification.
- 336 • The averaged change-score estimator is unbiased and stable but
- 337 less efficient, with the gap widening as the slope-to-noise ratio
- 338 grows.
- 339 • Misspecifying the random-slope variance inflates the GLS sam-
- 340 pling variability toward, and at high ratios beyond, that of the
- 341 simpler estimators.

342 To be completed by rgt.

343 *We simulate the factorial design described above under compound symmetry,*
344 *comparing the correctly specified GLS estimator, a GLS estimator that mis-*
345 *states the random-slope variance (it assumes $\sigma_b^2 = \sigma^2$ regardless of the truth),*
346 *and the averaged change-score estimator. For each cell we report the empir-*
347 *ical standard deviation of $\hat{\gamma}$ and the empirical power. The reduced replicate*
348 *count below is for render tractability; the manuscript value is 5000.*

349 6 Numerical examples

350 6.1 MIRIAD parameters

351 6.2 When to use which estimator

352 7 Discussion

Table 3: Monte Carlo comparison of GLS (correctly specified), GLS (misspecified σ_b^2 , assuming $\sigma_b^2 = \sigma^2$), and the averaged change-score estimator under compound symmetry (1000 replicates, $\Delta = 0.5$, two-sided $\alpha = 0.05$). Columns report the empirical standard deviation of $\hat{\gamma}$ and the empirical power.

Design	σ_b/σ	m	SD GLS	SD mis	SD avg	Pow GLS	Pow mis	Pow avg
(0,2,0)	1	50	0.238	0.238	0.375	0.537	0.537	0.495
(2,2,0)	1	50	0.210	0.210	0.625	0.677	0.677	0.206
(2,2,2)	1	50	0.151	0.151	0.802	0.902	0.902	0.192
(0,2,0)	1	100	0.169	0.169	0.266	0.823	0.823	0.769
(2,2,0)	1	100	0.146	0.146	0.462	0.926	0.926	0.393
(2,2,2)	1	100	0.108	0.108	0.584	0.997	0.997	0.345
(0,2,0)	5	50	1.012	1.012	1.525	0.083	0.653	0.083
(2,2,0)	5	50	0.241	0.528	3.064	0.555	0.600	0.067
(2,2,2)	5	50	0.164	0.284	4.020	0.873	0.765	0.054
(0,2,0)	5	100	0.726	0.726	1.089	0.108	0.733	0.109
(2,2,0)	5	100	0.162	0.360	2.081	0.857	0.748	0.067
(2,2,2)	5	100	0.106	0.189	2.752	0.997	0.936	0.060

Table 4: Sample size per group: GLS, ANCOVA, and averaged estimators (MIRIAD parameters, $\Delta = 0.25$, 90% power).

(J_0, J_1, J_2)	N GLS	N ANC	N Avg
(4,3,1)	36	480	8249
(4,3,0)	43	399	6702
(4,2,1)	60	399	6702
(2,3,1)	69	817	5328
(4,2,0)	78	339	5328
(2,3,0)	93	627	4109
(1,3,1)	99	1290	4156
(2,2,1)	104	627	4109
(0,3,1)	112	2190	2190
(1,2,1)	138	931	3100
(2,2,0)	145	478	3063
(0,2,1)	150	1462	1462
(4,1,1)	156	339	5328
(1,3,0)	185	931	3100
(2,1,1)	226	478	3063

[16] **GLS is the efficient choice under a correctly specified, adequately estimated model, which in practice means moderate-to-large samples.**

- GLS attains the minimum variance among linear unbiased estimators when the covariance is known.
- Its advantage is realized only when the variance components are estimated accurately, which requires sufficient sample size and follow-up.
- In small samples the instability of the estimated covariance can erase the theoretical gain.

To be completed by rgt.

The estimator comparison yields a graded recommendation rather than a single winner. Under correct specification GLS is efficient and dominates both ANCOVA and the averaged change-score estimator, as the ordering $\text{Var}_{\text{GLS}} \leq \text{Var}_{\text{ANC}} \leq \text{Var}_{\text{avg}}$ confirms. This dominance is a large-sample statement: it presumes the covariance components are known or well estimated. With moderate-to-large samples and a credible covariance model, GLS is the appropriate default.

[17] **When the variance components are uncertain, the simpler estimators are the safer choice because they do not amplify misspecification.**

- GLS weights observations by the inverse covariance, so errors in the assumed covariance propagate into the estimator.
- The averaged change-score estimator uses no variance components and is therefore unaffected by their misspecification.
- ANCOVA estimates a single covariate coefficient from the data, occupying a robust middle ground.

To be completed by rgt.

The misspecification analysis qualifies the case for GLS. Because GLS weights the data by an assumed inverse covariance, errors in the assumed G or R feed directly into the estimator, and the sandwich-variance calculations show that a mis-stated covariance can raise the GLS variance above that of the simpler estimators. The averaged change-score estimator is immune to this failure because it never uses the variance components. ANCOVA sits between the two: it estimates the coefficient on the run-in mean from the data rather than deriving it from assumed components, so it captures most of the run-in information while remaining comparatively robust. When the covariance is known only approximately, ANCOVA is the recommended default and the averaged estimator a conservative fallback.

[18] **The efficiency gap between GLS and the simpler estimators narrows as the design grows, so the choice matters most in small, information-poor designs.**

- With many run-in and treatment observations the phase means

become near-sufficient for the random slope.

- The averaged and ANCOVA estimators then recover most of the information GLS extracts, and the relative efficiency approaches one.
- The estimator choice is therefore most consequential when observations are few and the slope-to-noise ratio is high.

To be completed by rgt.

The practical importance of the estimator choice depends on the design. As the numbers of run-in and treatment observations grow, the phase means approach sufficiency for the subject-specific slope, the ANCOVA efficiency loss vanishes as $J_0 \rightarrow \infty$, and the relative efficiencies of all three estimators converge toward one. The gap is largest, and the choice of estimator therefore most consequential, in small designs with few observations per phase and a high slope-to-noise ratio, exactly the regime in which the estimated covariance is also least stable. This coincidence reinforces the preference for the simpler, more robust estimators in small studies.

[19] **Two extensions, the seemingly unrelated regression estimator and the constrained longitudinal data analysis estimator, would complete the efficiency picture.**

- The SUR formulation offers a middle ground between the averaged estimator and full GLS but adds implementation complexity.
- The cLDA estimator exploits the full covariance without specifying the random-effects distribution and is expected to rank between ANCOVA and GLS.
- A full comparison including both, and a feasible-GLS simulation under estimated components, is the natural continuation of this work.

To be completed by rgt.

Two estimators not fully developed here would round out the comparison. The seemingly unrelated regression formulation of Section 2.4 treats the run-in and treatment periods as linked equations and offers a compromise between the averaged estimator and full GLS, at the cost of additional implementation complexity; its variance derivation remains to be completed. The constrained longitudinal data analysis estimator uses the full covariance without committing to a random-effects distribution and should fall between ANCOVA and GLS in efficiency. A comprehensive treatment would compare all of these under a feasible-GLS simulation in which the variance components are estimated rather than assumed, which is the setting in which the robustness distinctions drawn here become decisive.

8 Conflict of interest

The author declares no conflicts of interest.

438 9 Funding

439 This work received no specific funding.

440 10 Data availability statement

441 [20] **The package implementing all three estimators, the sim-**
442 **ulation drivers, and the shared bibliography are openly avail-**
443 **able in the project repository.**

- 444 • The estimator implementations reside in the package source files.
- 445 • The simulation drivers and shared compendium bibliography are
- 446 included.
- 447 • Specific paths appear in the Reproducibility section below.

448 *To be completed by rgt.*

449 *The ‘runinpower’ R package implementing the GLS, ANCOVA, and averaged-*
450 *change-score estimators (‘ancova.R’, ‘averaged.R’), the simulation drivers,*
451 *and the shared bibliography across the four-paper compendium are openly*
452 *available in the project repository. Specific paths are listed in the Repro-*
453 *ducibility section below.*

454 11 Reproducibility

455 [21] **The manuscript was rendered with R 4.4.x using the**
456 **package source and a small set of standard analysis and build**
457 **packages.**

- 458 • The in-package source supplies the estimator implementations.
- 459 • The GLS comparator in the simulation relies on nlme, with
- 460 tinytest as the testing harness.
- 461 • The manuscript is built with rmarkdown and bookdown.

462 *To be completed by rgt.*

463 *This manuscript was rendered with R 4.4.x. Principal packages used are*
464 *the in-package ‘runinpower’ source, ‘nlme’ (the GLS fitter for the simulation*
465 *comparator), ‘tinytest’ (testing harness), and ‘rmarkdown’ plus ‘bookdown’*
466 *for the manuscript build.*

467 [22] **The simulation uses a fixed master seed with determin-**
468 **istic per-replicate offsets to ensure reproducibility.**

- 469 • A single master seed is set at the entry point of the Monte Carlo
- 470 comparison.
- 471 • Per-replicate seeds derive deterministically from the master seed
- 472 by adding the replicate index.
- 473 • This scheme makes the full simulation exactly reproducible.

474 *To be completed by rgt.*

475 ***Random-number generator.** The Monte Carlo simulation comparing the*
476 *three estimators sets 'set.seed(20260418)' at its entry point. Per-replicate*
477 *seeds derive from the master seed by '+ rep_idx'.*

478 [23] **All code, drivers, companion manuscripts, and the shared**
479 **bibliography are located at documented paths within the**
480 **compendium.**

- 481 • The package source and simulation drivers reside in their respec-
482 tive directories.
- 483 • The three companion manuscripts are stored alongside this paper
484 in the compendium.
- 485 • The manuscript source and shared bibliography are co-located in
486 the paper directory.

487 *To be completed by rgt.*

488 ***Data and code availability.** The R package source is at 'R/'. Simula-*
489 *tion drivers are at 'analysis/scripts/'. The companion manuscripts (Papers*
490 *1, 2, 3) are at 'analysis/report/01,02,03-*/'. The manuscript source and*
491 *shared bibliography ('references.bib', a symlink to '../..references.bib') are at*
492 *'analysis/report/04-gls-vs-ttest/'.*

493 12 References

494

495 *Rendered on 2026-06-23 at 18:51 PDT.*

496 *Source: /Users/zenn/Library/CloudStorage/Dropbox/prj/res/04-runin-power-analysis/runinpower/a*